Website — Software Topology

BROWSER

User's Web Browser

Chrome · Firefox · Safari

DISCORD

Webhook target

discord.com

HTTPS · static assets

HTTPS / WSS

NETLIFY

Static Hosting · CDN

index.html
blog.html · blog.js
irc.html · irc.js
about.html · profile pages
ControlSystems.css · .js · grid-fx.js

Netlify Functions

transmit.js
POST /.netlify/functions/transmit
Contact form → Discord webhook
Env: DISCORD_WEBHOOK_URL

CLOUDFLARE NETWORK

Zero Trust · Named tunnel · Two public hostnames

cloudflared agent
Maintains outbound tunnel
to Cloudflare edge

api.your-domain.com
→ localhost:3000
GET /api/posts
POST /api/subscribe
GET /admin (localhost only)

irc.your-domain.com
→ localhost:8097
WebSocket (WSS)
#control-systems
Ergo IRCd

POST webhook · contact form → Discord

Tunnel
(outbound)

routes

routes

NixOS HOST SERVER

NixOS 25.11 (flake-based) · systemd services · Cloudflare Tunnel (named)

ddl-cloudflared

cloudflared tunnel run
pkgs.cloudflared
DynamicUser=true
Restart=on-failure (10s)
Env: TUNNEL_TOKEN
/etc/cloudflared/tunnel.env
After=network-online.target

systemd: ddl-cloudflared

ddl-backend

Node.js 22 · Express · PORT 3000
pkgs.nodejs_22
User=helios · ProtectSystem=strict
Restart=on-failure (5s)
Env: PORT · ENCRYPTION_KEY
ADMIN_PASSWORD
ALLOWED_ORIGIN

systemd: ddl-backend

ddl-ergo

Ergo IRCd · Go binary
/opt/ergo/ergo
User=ergo (system user)
WebSocket 127.0.0.1:8097
Open guest access · NickServ
History: 512 · replay 50 on join
IP cloaking · SASL optional

systemd: ddl-ergo

SQL

POST /sub

SQLite (better-sqlite3)

backend/data/posts.db
Blog posts · Admin CRUD
ReadWritePaths: data/ uploads/
Gitignored (posts.db)

better-sqlite3 · synchronous writes

Encrypted Email Store

backend/data/subscribers.json
AES-256-GCM · {iv, tag, ciphertext}
Env: ENCRYPTION_KEY (64 hex chars)

LEGEND

- Live data flow (HTTPS / WSS)
- Serverless / webhook (dashed)
- Cloudflare Tunnel (outbound)
- IRC / WebSocket route
- Internal / storage (dashed)